

Executive Summary

Meeting of ITU-T SG17 'Security', Geneva, 23 August - 2 September 2022

Hot topics

- Intelligent transport system security
- Revised TTCN-3 standards
- 5G security
- Cloud security
- DLT based security services
- IoT security

1 Meeting Output (meeting statistics see Annex E below)

- Output standards (29, see Annex A):
 - TAP approval (3): Details are in Annex A a).
 - TAP determined (11): 11 new Recommendations. Details are in Annex A b).
 - AAP consented (13): 12 revised and one new Recommendations for AAP Last Call. Details are in Annex A c).
 - Agreed (2): 2 new Supplements. Details are in Annex A d).
- New work items (9, see Annex C).
- Work item discontinued (0).
- [ITU Workshop on "Security for 5G and beyond"](#) was held successfully on 22 August 2022
- New national OID registration authority for Kenya: {joint-iso-itu-t(2) country(16) ke(404)}
- A.5 qualification of SIA (Secure Identity Alliance)
- 4 SG17 specific TD templates (Q meeting agenda; Q, WP and CG meeting reports)
- Replacement of SG17 Vice-Chair from Argentina
- SG17 subgroups
 - JCAs:
 - JCA-COP: terminated
 - [JCA-IdM](#): continued with revised ToR, and held its 30th meeting on Friday 26 August 2022
 - [Correspondence Groups](#)
 - CG-SG17-wtsa24-prep: new (ToR in [TD567](#))
 - CG-secapa (Correspondence Group on Security Capability and Architecture): continued with revised ToR
 - CG-SG17-meeting (Correspondence Group on effective SG17 meeting): terminated
 - Advisory group on X.509: new (ToR in [TD639](#))

2 Next SG17 meetings

2.1 3rd SG17 meeting: Geneva, Tuesday 21 February - Friday 3 March 2023 (9 working days, physical meeting with remote participation)

- Joint ITU-T SG17/ISO TC 307 workshop on "DLT security, identity management and privacy", Monday 20 February 2023, 09:30-17:30
- Open and extended management team meeting on Monday 20 February 2023, 19:00-21:00.
- 31st JCA-IdM meeting on Friday 24 February 2023, 14:30-16:00
- 25 Candidate texts for action, details see Annex B.

2.2 4th SG17 meeting: Korea, exact dates and venue to be decided by Dec 2022

2.3 Interim RGMs

8 Questions plan to hold the following 9 RGMs before next SG17 meeting:

#	Q	Date	Place/Host	Subject/objective
1.	2/17	8 Dec 2022	e-meeting	• prepare texts for action in next SG17 meeting: X.5Gsec-message. • review all work items and identify future topics for Q2/17.
2.	3/17	20 Sept 2022	e-meeting	• X.1051rev2
3.	3/17	3 rd week Nov 2022	e-meeting	• X.sup-cdc
4.	4/17	19 Jan 2023	e-meeting	• X.stie and X.taeii
5.	8/17	13-14 Jan, 2023	e-meeting	• progress Q8 work items that are planned for approval or determination in the next SG17 meeting
6.	10/17	Nov 2022	e-meeting	• progress all the work of Q10/17
7.	11/17	12-13 Dec 2022	Luxembourg	• Progression of X.pki-em (Public-key infrastructure: Establishment and maintenance) • Progression of X.510 (Information technology – Open Systems Interconnection – The Directory: Protocol specifications for secure operations)

8.	13/17	24-25 Nov 2022	e-meeting	address all work items and future topic.
9.	15/177	Nov 2022	Singapore	progress all the work of Q15/17

Annex A
Actions taken on Recommendations, and other texts at SG17 closing
plenary on 2 September 2022

a) TAP Recommendations approved (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor (s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	2/17	X.1813 (X.5G-vs)	Security and monitoring requirements for operation of vertical services supporting ultra-reliable and low latency communication (URLLC) in IMT-2020 private network	New	Jae Eon Oh, Seongki Shin, Heung Youl Youm	TD554		2020-09	2022-05
2.	2/17	X.1814 (X.5Gsec-guide)	Security guidelines for IMT-2020 communication system	New	Mee Yeon Kim, Keundug Park, Heung Youl Youm	TD351		2019-01	2022-05
3.	6/17	X.1352 (X.iotsec-4)	Security requirements for IoT device and gateway	New	Jiho Bang, Wonsuk Chung, Sanggeol Lee, Hosoek Ryu	TD549		2018-09	2022-05

b) TAP Recommendations determined (WTSA-20 Resolution 1)

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	2/17	X.1815 (X.5Gsec-ecs)	Security guidelines and requirements for IMT-2020 edge computing services	New	Feng Gao, Jae Hoon Nah, Junjie Xia, Bo Yu, Xiaojun zhuang	TD578 A.5 in TD605		2019-01	2022-09
2	2/17	X.1816 (X.5Gsec-ssl)	Guidelines and requirements for classifying security capabilities in IMT-2020 network slice	New	Ke Wang, Li Su, Bo Yang, Zhiyuan Hu	TD555 A.5 in TD552		2020-09	2022-09
3	6/17	X.1353 (X.ztd-iot)	Security methodology for zero-touch deployment in massive IoT based on blockchain	New	Xin Kang, Haiguang Wang, Weidong Wang	TD584		2020-09	2022-09
4	7/17	X.1454 (X.sles)	Security measures for location enabled smart office service	New	Hao Dong, Feng Gao, Jae Hoon Nah, Junjie Xia	TD576		2019-09	2022-09
5	8/17, (14/17)	X.1411 (X.BaaS-sec)	Guidelines on blockchain as a service (BaaS) security	New	Nan Meng, Kyeong Hee Oh, Zixiang Wang, Xuan Zha	TD622		2019-09	2022-09
6	8/17	X.1644 (X.sgdc)	Security guidelines for distributed cloud	New	Ye Tao, Lei Xu, Mark McFadden, Laifu Wang	TD618		2019-01	2022-09
7	13/17	X.1380 (X.edr-sec)	Security guidelines for cloud-based data recorders	New	Sang-Woo Lee, Seungwook Park	TD596		2018-03	2022-09

			in automotive environments						
8	13/17	X.1381 (X.eivn-sec)	Security guidelines for Ethernet-based In-Vehicle networks	New	Sang-Woo Lee, You-Sik Lee	TD613		2018-09	2022-09
9	13/17	X.1382 (X.fstiscv)	Guidelines for sharing security threat information on connected vehicles	New	Min Shu, Wenlei Wang, Yunwei Zhao, Xiaochun Yun	TD614 A.5 TD510		2018-09	2022-09
10	13/17	X.1383 (X.srzd)	Security requirements for categorized data in vehicle-to-everything (V2X) communication	New	Yaping Sun, Huirong Tian, Nan Meng, Takamasa Isohara	TD603		2018-09	2022-09
11	14/17	X.1410 (X.sa-dsm)	Security architecture for data-sharing management based on the distributed ledger technology	New	Feng Gao, Zhiyuan Hu, Min Shu, Yunwei Zhao	TD553			2022-09

c) AAP Recommendations consented (Recommendation ITU-T A.8)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1	11/17	Z.161	Testing and Test Control Notation version 3: TTCN-3 core language	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 201 873-1	2022-09	2022-09
2	11/17	Z.161.1	Testing and Test Control Notation version 3: TTCN-3 language extensions: Support of interfaces with	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 786	2022-09	2022-09

			continuous signals						
3	11/17	Z.161.2	Testing and Test Control Notation version 3: TTCN-3 language extensions: Configuration and deployment support	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 781	2022-09	2022-09
4	11/17	Z.161.3	Testing and Test Control Notation version 3: TTCN-3 language extensions: Advanced parameterization	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 784	2022-09	2022-09
5	11/17	Z.161.4	Testing and Test Control Notation version 3: TTCN-3 language extensions: Behaviour types	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 785	2022-09	2022-09
6	11/17	Z.161.5	Testing and Test Control Notation version 3: TTCN-3 language extensions: Performance and real time testing	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 782	2022-09	2022-09
7	11/17	Z.161.6	Testing and Test Control Notation version 3: TTCN-3 language extensions: Advanced Matching	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 203 202	2022-09	2022-09

8	11/17	Z.161.7	Testing and Test Control Notation version 3: TTCN-3 Language Extensions: Object-Oriented Features	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 230 790	2022-09	2022-09
9	11/17	Z.165	Testing and Test Control Notation version 3: TTCN-3 runtime interface (TRI)	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 201 873-5	2022-09	2022-09
10	11/17	Z.165.1	Testing and Test Control Notation version 3: TTCN-3 extension package: Extended TRI	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 202 789	2022-09	2022-09
11	11/17	Z.166	Testing and Test Control Notation version 3: TTCN-3 control interface (TCI)	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 201 873-6	2022-09	2022-09
12	11/17	Z.167	Testing and Test Control Notation version 3: Using ASN.1 with TTCN-3	Rev	Dieter Hogrefe	TD515 A.5 in TD516	ETSI ES 201 873-7	2022-09	2022-09
13	13/17	X.1377 (X.ipscv)	Guidelines for an intrusion prevention system for connected vehicles	New	Seonghoon Jeong, HuyKang Kim, Sang-Woo Lee, Seungwook Park	TD593		2019-09	2022-09

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) agreed

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	4/17	X.Suppl.37 (X.sup.cs-ML)	Supplement to X.1231: Countering spam based	New	Wei Liu, Ye Tao	TD563		2019-09	2022-09

			on machine learning						
2.	7/17	X.Suppl.38 (X.Sup.cta)	Supplement to X.1152: Use cases for contact tracing applications to prevent spread of infectious diseases	New	Mee Yeon Kim, Sungchae Park, Heung Youl Youm	TD642		2020-09	2022-09

Annex B

Recommendations planned for action in SG17 Feb/Mar 2023 meeting

a) TAP Recommendations planned for TAP approval (WTSA-20 Resolution 1)

11 determined texts as shown in b) of Annex A.

b) TAP Recommendations planned for TAP determination (WTSA-20 Resolution 1)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing
1.	2/17	X.5Gsec-message	Security requirements for 5G message service	New	Hang Dong, Le Yu, Hongyang Zhang	TD607		2021-04	2023-03
2.	4/17	X.stie	Structured Threat Information Expression	New	Michael Rosa, Duncan Sparrell	TD546	OASIS STIX Version 2.1	2022-09	2023-03
3.	4/17	X.taeii	Trusted Automated Exchange of Intelligence Informa	New	Michael Rosa, Duncan Sparrell	TD546	OASIS TAXII Version 2.1	2022-09	2023-03
4.	6/17	X.sc-iot	Security Controls for Internet of	New	Liu Lijun, Koji Nakao	TD616		2018-09	2023-03

			Things (IoT) system						
5.	7/17	X.rdda	Requirements for data de-identification assurance	New	Feng Gao, I Seok Kang, Soonseok Kim, Jihun Kim, Byunghoon Lee, Hyungjin Lim, Yunsik Park, Heung Youl Youm	TD659		2019-01	2023-03
6.	8/17	X.nssa-cc	Requirements of network security situational awareness platform for cloud computing	New	Maofei Chen, Huamin Jin, Zhaoji Lin, Laifu Wang, Yi Zhang	TD595		2019-09	2023-03

Notes:

(1) In case of joint Question activity, the lead Question is given without parentheses and other Questions are shown in parentheses; such entries are only shown in the table against the lead Question.

c) AAP Recommendations planned for AAP consent (Recommendation ITU-T A.8)

#	Q ⁽¹⁾	Acronym	Title	New / Revised	Editor(s)	Location of Text	Equivalent e.g., ISO/IEC	Start of work	Timing

1.	3/17	X.1051-rev2	Information security, cybersecurity and privacy protection - Information security controls based on ISO/IEC 27002 for telecommunications organizations	Rev	Hang Dong, Jinghua Min, Thaib Mustafa	TD544	ISO/IEC 27011	2020-03	2023-03
2.	4/17	X.arc-ev	Functional requirements for a secured process to evaluate technical vulnerabilities	New	Wei Li, Shan Xue, Chen Zhang	TD571		2019-08	2023-03
3.	7/17	X.websec-7	Reference monitor for online analytics services	New	Hyungjin Lim, Jah Hoon Nah, Jongyoul Park, Junjie Xia	TD573		2014-09	2023-03
4.	11/17	X.pki-em	Public-key infrastructure: Establishment and maintenance	New	Erik Andersen	TD671		2012-08	2023-03
5.	11/17	X.510 Amd.1	Information technology – Open Systems Interconnection – The Directory: Protocol specifications for secure operations	New	Erik Andersen	TD670		2020-08	2023-03
6.	14/17	X.srscm-dlt	Security Requirements for Smart Contract Management based on distributed ledger technology	New	Tietao Guo, Li Li, Kyeong Hee Oh, Min Shu, Yang Wu	TD558		2020-09	2023-03

d) Non-normative texts (Technical Report, Supplement, Implementers' Guide, etc) planned for agreement

#	Q	Acronym	Title	New / Revised	Editor(s)	Location of text	Start of work	Timing
1.	3/17	X.sup-cdc	Supplement to X.1060: X.1060 Tutorial material	Rev	Shigenori Takei, Hiroshi Takechi	TD653	2022-09	2023-03
2.	6/17	TR.ibc-cd	Technical report: Guideline for identity-based cryptosystems used for cross-domain secure communications	New	Fuwen Liu, Li Su, Junzhi Yan, Bo Yang	TD583	2021-04	2023-03

Annex C New work items

The following new work items were agreed to be added to the SG17 Work Programme:

#	Question	NWI	Approval	TD	Title	C
1.	2/17	X.5Gsec-ctrl	TAP	TD577	Security controls for operation and maintenance of 5G network systems	C176
2.	3/17	X.sup-cdc	Agreement	TD653	Supplement to X.1060: X.1060 Tutorial material	-
3.	4/17	X.stie	TAP	TD546	OASIS STIX Version 2.1	TD546
4.	4/17	X.taeii	TAP	TD546	OASIS TAXII Version 2.1	TD546
5.	8/17	X.sg-scmr	AAP	TD620	Security guidelines for selecting computing methods and resources from Cloud Service Providers	C192

6.	11/17	XSTR.x509ac4sc	Agreement	TD638	Technical Report: A use case of X.509 Attribute Certificate for Supply Chain	C172
7.	13/17	X.sup.cv2x-sec	Agreement	TD570	Supplement to X.1813: Security deployment models and requirements for the operation of C-V2X services supporting ultra-reliable and low latency communication (URLLC)	C131
8.	13/17	X.evpnc-sec	TAP	TD590	Security guidelines for electric vehicle plug and charge (PnC) service using vehicle identity (VID)	C152
9.	15/17	X.sec_QKDNI	AAP	TD588	Security requirements for Quantum Key Distribution Network interworking (QKDNI)	C126

Annex D Work items discontinued

Question	Acronym	Title
None		

Annex E SG17 meeting Statistics

Participants ([TD333R1](#))

	participants	countries	Member States	Sector Members	SG17 Associates	Academia	Invited Experts
Announced	292	52					
Final	282	45	39	30	1	4	7

Meeting input and organization

Table of SG17 statistics of this and some past meetings

	2022-09	2022-05*	2021-09*	2021-04*	2020-09*	2020-03*	2019-09	2019-01	2018-09	2018-03	2017-09	2017-03	2016-09	2016-03	2015-09	2015-04
C	104	101	98	104	110	121	151	118	144	113	106	78	81	66	74	80
LS/i	55	72	46	57	58	49	67	58	47	40	45	43	38	47	56	60
LS/o	20	20	26	17	28	29	40	39	37	38	46	38	23	32	42	42
TD	344	331	336	307	434	438	508	380	420	395	426	368	391	418	371	386

Note * - fully virtual meeting

Contributions: 104 – stable, DDP: 98%.

- APT 94 (90%) [China 42.5 + Korea 38.5 + Japan 10 + India 2 + Singapore 1]
- Americas 7 (7%) [US 4]
- AFR 2 (2%) [Comoros, Mali & Senegal]
- EUR 1 (1%) [Denmark 1]
- RCC (0), ARAB (0), LAM (0)

LS: matrix in TD352R1

- incoming 55 - stable
- Outgoing 20 - stable

TDs: 344 - stable